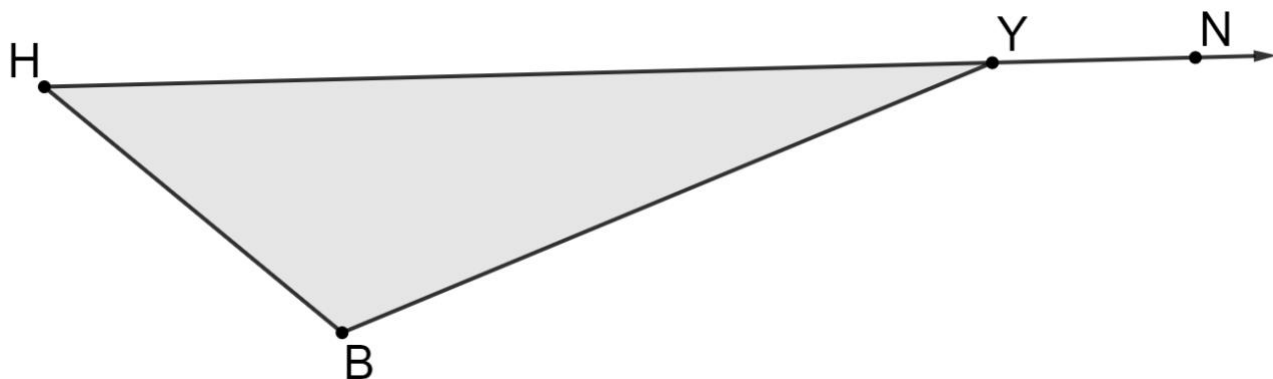


1. Triangle BHY is shown, where $m\angle BHY = (2x + 7)^\circ$, $m\angle HBY = (8x - 18)^\circ$, and $m\angle NYB = (4x + 91)^\circ$.



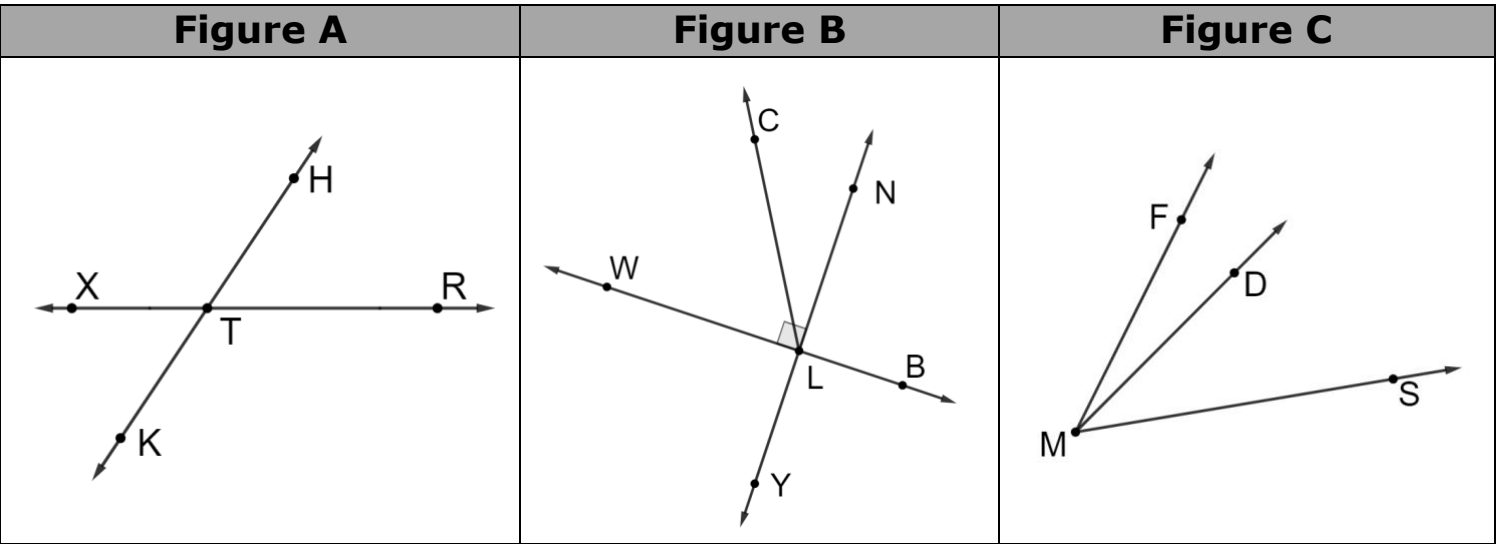
- a. Determine the value of x . Show your work.

work	
$x =$	17

- b. Determine measure of angle NYB in degrees. Show your work.

work	
$m\angle NYB$	159°

2. Three figures are shown.



- a. In figure A, name an angle that is congruent to $\angle XTK$.

b. Name a pair of complementary angles.

c. In figure C, what type of angle are $\angle DMF$ and $\angle DMS$?

d. In figure B, name an angle that is supplementary to $\angle NLB$.

$\angle HTR$
$\angle WLC$ and $\angle CLN$
adjacent angles
$\angle BLY$, $\angle WLN$, or $\angle WLY$

3. The measures of the interior angles of $\triangle HTR$ are shown.

$m\angle THR = (4x)^\circ$

$m\angle RTH = (16x)^\circ$

$m\angle HRT = (5x)^\circ$

Determine the measure, in degrees, of $\angle HRT$. Show your work.

work	
$m\angle HRT$	36°

4. Angles MVH and HVK are adjacent complementary angles. The measure of angle HVK is $(6.25x - 10)^\circ$ and the measure of angle MVH is $(2x + 7.6)^\circ$.

a. Determine the value of x . Show your work.

work	
$x =$	11.2

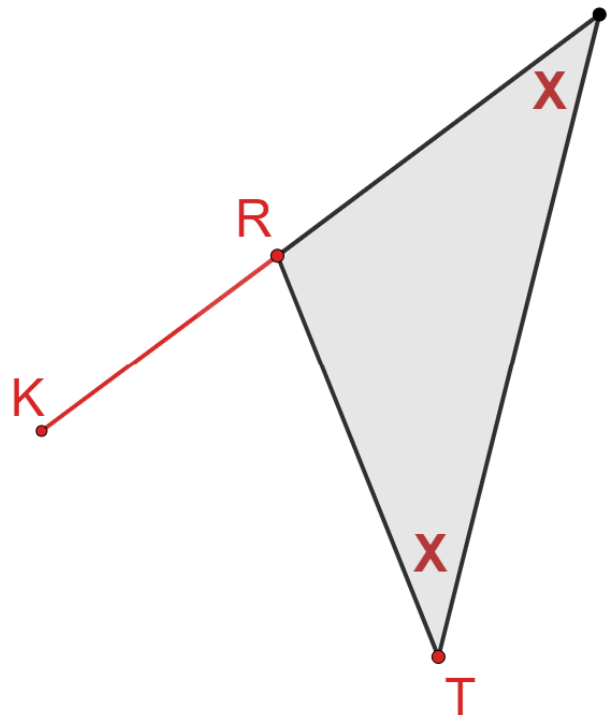
b. Determine measure of angle HVK in degrees. Show your work.

work	
$m\angle HVK$	60°

5. A triangle is shown.

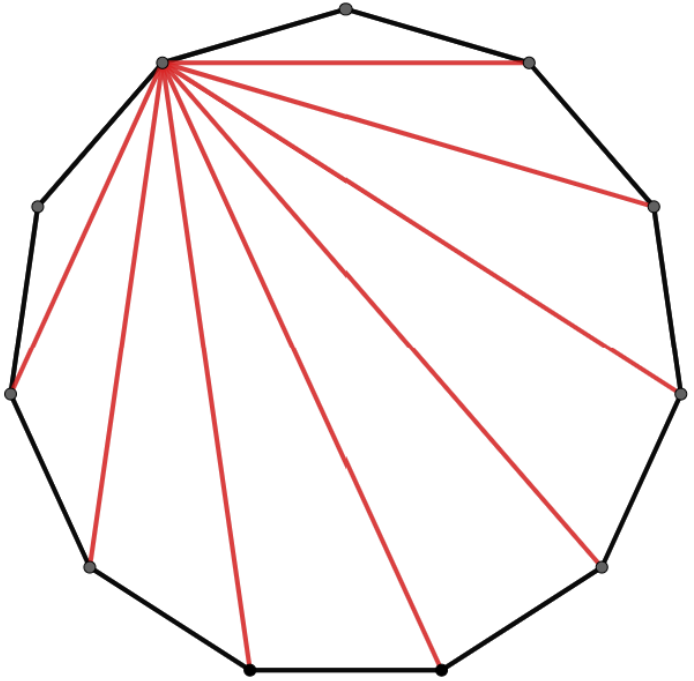
a. Draw an exterior angle of the triangle and name it $\angle KRT$.

b. Place an **X** inside the angles whose sum is equal to $m\angle KRT$.



6. A regular polygon is shown.

- a. Draw the triangles that can be used to determine the sum of the interior angles of the polygon.



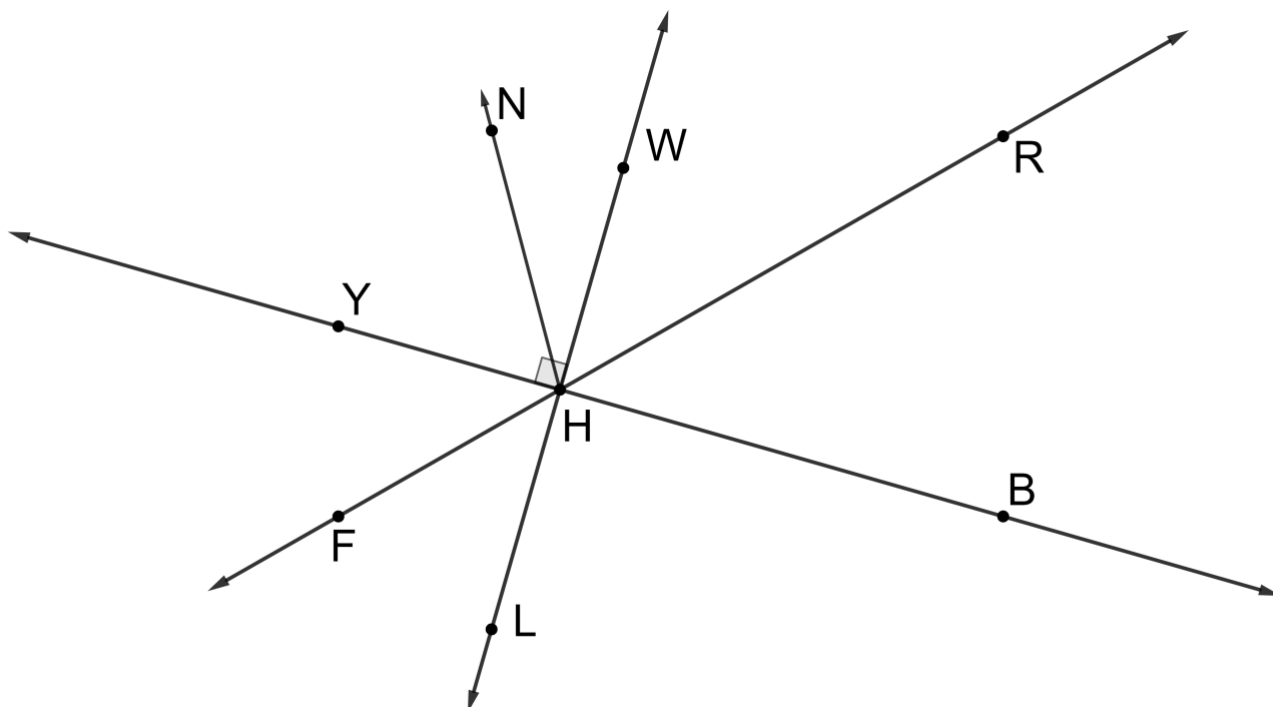
- b. Determine the sum of the interior angles in the polygon.

work	
Sum	1620°

- c. Determine the measure of one of the interior angles in the polygon.

work	
Measure of One Interior Angle	$147.\overline{27}^\circ$

7. A figure is shown where $m\angle YHW = 90^\circ$.



a. Name an angle that is supplementary to $\angle BHF$.

$\angle FHY$ or $\angle BHR$

b. Name an angle that is the vertical angle of $\angle YHF$.

$\angle BHR$

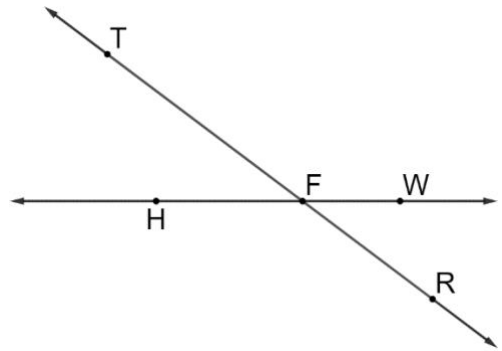
c. If $m\angle RHB = 46^\circ$, then find $m\angle FHB$.

134°

8. Determine the sum, in degrees, of the interior angles in a regular polygon with 21 sides.

work	
Sum of Interior Angles	3420°

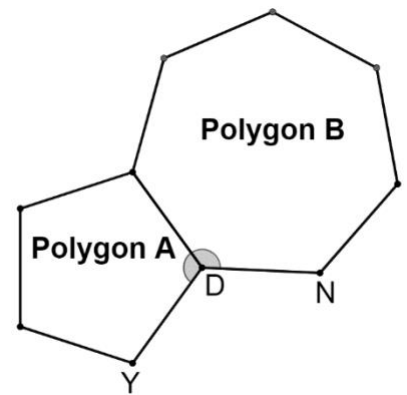
9. Lines TR and HW are shown, where $m\angle HFT = (5x - 7.5)^\circ$ and $m\angle HFR = (15x + 13.5)^\circ$.



Determine $m\angle WFR$, in degrees. Show your work.

work	
$m\angle WFR$	36°

10. Two polygons are shown, where polygon A has 5 sides and polygon B has 7 sides. Angle YDN is marked, as shown.



Determine the measure, in degrees, of angle YDN . Round to the nearest thousandth.
Show your work.

work	
$m\angle YDN$	236.571°